

Question 1 (12 marks)**Marks**

- (a) Evaluate $\sqrt[3]{\frac{4.37+0.8}{1.2-0.9}}$, correct to 2 decimal places. **1**
- (b) Express $(4x)^{-\frac{1}{2}}$ without a fractional index. **1**
- (c) Simplify completely:
- (i) $(x + y) - (x - y)$ **1**
- (ii) $\frac{m-1}{3} - \frac{2m+5}{2}$ **2**
- (iii) $3\sqrt{54} + 2\sqrt{24}$ **2**
- (d) Write $0.03\overline{4}$ as a rational number in the form of $\frac{p}{q}$, where **2**
p and q are integers.
- (e) Express $\frac{3\sqrt{5}}{\sqrt{5}-1}$ with a rational denominator. **2**
- (f) Factorise completely $8a^3 - 27$ **1**

Question 2 (12 Marks) Start this question on a new page. **Marks**

(a) Show that $f(x) = -\frac{4}{x}$ is an odd function **2**

(b) Solve $|1 + 5x| \geq 4$ and graph your solution on the number line **3**

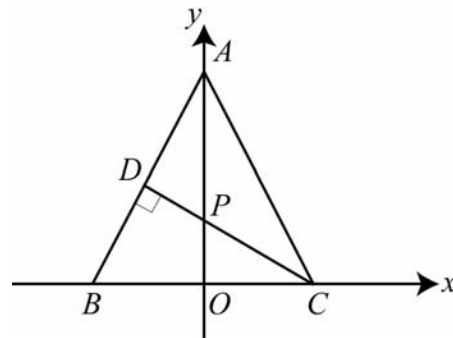
(c) State the domain and range of the function $y = -\sqrt{9 - x^2}$ **2**

(d) Sketch the region where **3**

$$y > -3, \quad y - x - 1 \leq 0 \quad \text{and} \quad y \leq -x + 1$$

hold simultaneously. Points of intersection are not required.

(e) Find the value(s) of x for which $f(x) = 0$ when $f(x) = x^2 - 3x - 1$ **2**
Give your answer in exact form.

Question 3 (12 Marks) Start this question on a new page.**Marks**

NOT TO SCALE

In the diagram, A is the point $(0, 4)$ and B is $(-3, 0)$. CD is perpendicular to AB .

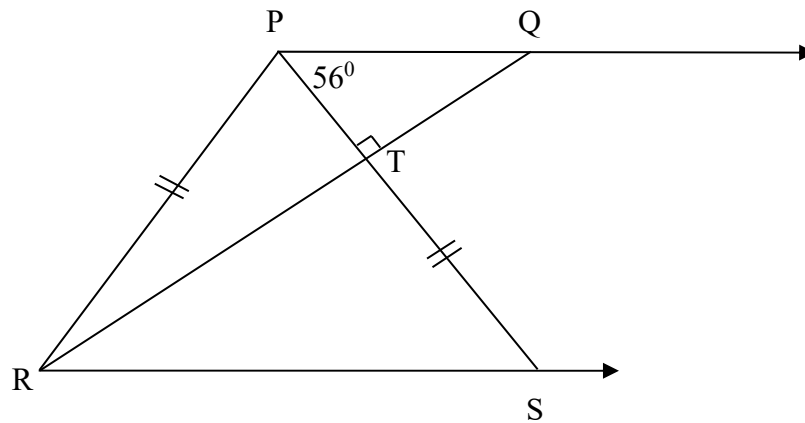
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|-----|---|---|
| (a) | Find the length of AB . | 1 |
| (b) | If $AB = BC$, find the coordinates of C . | 1 |
| (c) | If $CD \perp AB$, show that the equation of CD is $3x + 4y = 6$. | 3 |
| (d) | CD intersects the y -axis in P . Show that $CP = \frac{5}{2}$. | 2 |
| (e) | Prove that $\triangle ADP \equiv \triangle COP$. | 3 |
| (f) | Calculate the area of the quadrilateral $DPOB$. | 2 |

Question 4 (12 Marks) *Start this question on a new page.* **Marks**

- (a) Given $\cos \theta = \frac{12}{13}$, find the exact value of $\tan \theta$ if θ is an acute angle. **2**
- (b) Solve for $0^\circ \leq \theta \leq 360^\circ$ if $2 \sin \theta + 1 = 0$ **2**
- (c) Simplify $\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x}$ **3**
- (d) At 14:00 hours, a ship is at point A, which is due south of a small island, I. The ship is travelling on a bearing of 020° at 15 km/h. After 1 hour and 40 minutes, it is at a point B, where it has a bearing of 135° from the island.
- (i) Draw a neat diagram indicating all the relevant information. **1**
- (ii) Calculate the distance from A to I to 2 significant figures. **2**
- (iii) Find the time, to the nearest minute, at which the island will be due west of the ship. **2**

Question 5 (8 Marks) Start this question on a new page.**Marks**

(a)



In the diagram above, $PQ \parallel RS$, $PR = PS$,
 $\angle QPS = 56^\circ$ and $PT \perp RQ$

(i) Find $\angle PRS$. No reasoning is required.

1

(ii) Show that $\angle PRQ = 22^\circ$

2

(b) For a regular octagon,

find (i) the sum of the interior angles

1

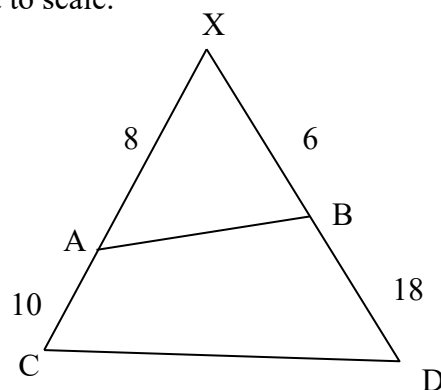
(ii) the size of each interior angle.

1

(c) In the figure below, prove $\triangle XAB \parallel \triangle XDC$.

3

The figure is not to scale.



Question 6 (16 Marks) Start this question on a new page. **Marks**

(a) Differentiate

(i) $y = 2x^3 + 5x - \frac{1}{x}$ **2**

(ii) $y = (1 - 4x^5)^3$ **2**

(iii) $y = 4x\sqrt{x}$ **2**

(iv) $y = \frac{6x + 5}{1 - 2x}$ **2**

(b) Find the value of $\lim_{x \rightarrow \infty} \frac{4x^2 + x}{1 + 2x - x^2}$ **2**

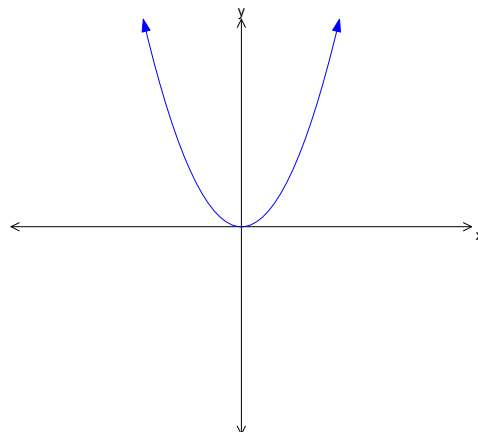
(c) If $V = 4t^2 - 12t + 2$,

find (i) $\frac{dv}{dt}$ **1**

(ii) the value of t when $\frac{dv}{dt} = 0$ **1**

(d) Find the equation of the tangent to the curve $y = \frac{2}{x + 1}$ **3**
at the point where $x = 3$.

(e) Copy this graph onto your paper. On the same set of axes, draw a sketch of the gradient function. **1**



Question 7 (8 Marks) Start this question on a new page. **Marks**

(a) For the quadratic polynomial $y = -2x^2 + 7x + 4$,

find (i) the roots by factorisation **2**

(ii) the axis of symmetry **1**

(iii) the vertex **1**

(iv) Hence or otherwise, sketch the curve **1**

$y = -2x^2 + 7x + 4$, showing all the details
above and the y intercept.

(b) Find the value(s) of k such that $x^2 + (k + 1)x + 4 = 0$ **3**
has no real roots.

END OF PAPER

QUESTION 1 (12 MARKS)

(a) 2.58

1

(b) $(4x)^{-\frac{1}{2}} = \frac{1}{\sqrt{4x}}$

$= \frac{1}{2\sqrt{x}}$

1

(c) (i) $x+y-x+y$
 $= 2y$

1

(ii) $\frac{2(m-1) - 3(2m+5)}{6}$

$= \frac{2m-2-6m-15}{6}$

1

$= \frac{-4m-17}{6}$

1

(iii) $3\sqrt{9}\sqrt{6} + 2\sqrt{4}\sqrt{6}$
 $= 9\sqrt{6} + 4\sqrt{6}$
 $= 13\sqrt{6}$

1

1

(d) $x = 0.0344...$

$10x = 0.3444...$

$9x = 0.31$ $x = \frac{0.31}{9}$

1

$x = \frac{31}{900}$

1

(e) $(2a-3)(4a^2+6a+9)$

1

QUESTION 2 (12 MARKS)

(a) $f(-a) = -\frac{-4}{a} = \frac{4}{a}$

1

$-f(a) = -\frac{4}{a} = \frac{4}{a}$

1

Since $f(-a) = -f(a)$, then function is ODD

(b) $1+5x \geq 4$ or $-(1+5x) \geq 4$

$5x \geq 3$

$x \geq \frac{3}{5}$

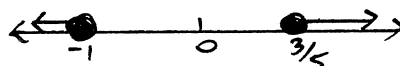
1

$-1-5x \geq 4$

$-5x \geq 5$

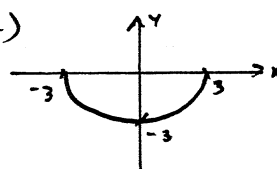
$x \leq -1$

1



1

(c)



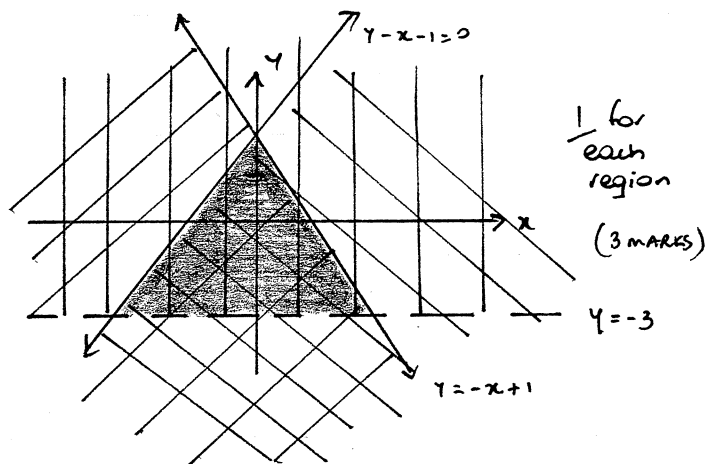
D: $-3 \leq x \leq 3$

1

R: $-3 \leq y \leq 0$

1

(d)



(e) $x = \frac{3 \pm \sqrt{9 - 4 \cdot 1 \cdot 1}}{2}$

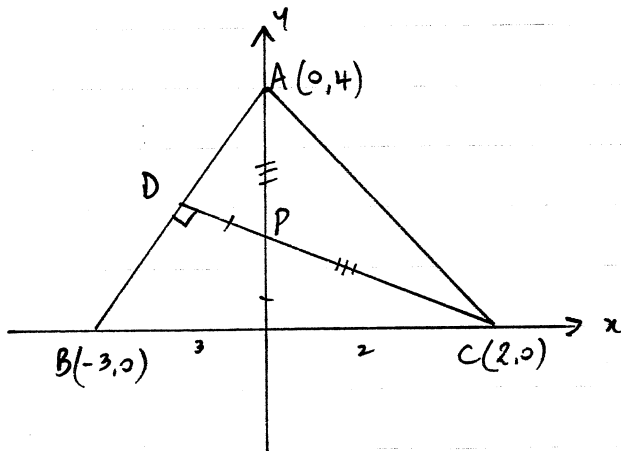
1

$= \frac{3 + \sqrt{5}}{2}$ or $\frac{3 - \sqrt{5}}{2}$

1

QUESTION 3

(12 MARKS)



(a) $AB = \sqrt{3^2 + 4^2}$
 $= 5 \text{ units}$ $\perp$

(b) $C(2,0)$ $\perp$

(c) $\text{grad } AB = \frac{4}{3}$
 $\text{grad } DC = -\frac{3}{4}$ $\perp$

Eqn CD: $\frac{y}{x-2} = -\frac{3}{4}$ $\perp$

$4y = -3x + 6$
 $3x + 4y = 6$ $\perp$

(d) coords of P:
 let $x=0$, $4y=6$
 $y = \frac{3}{2}$
 $\therefore P(0, \frac{3}{2})$ $\perp$

CP = $\sqrt{2^2 + (\frac{3}{2})^2}$
 $= \sqrt{4 + \frac{9}{4}}$
 $= \sqrt{\frac{25}{4}}$ $\perp$
 $= \frac{5}{2} \text{ units}$

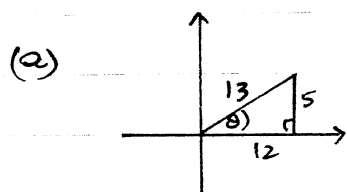
(e) $\angle DPA = \angle OPC$ (vert. opp $\angle$ s =) $\perp$
 $\angle ADP = \angle COP$ (given - both 90°) $\perp$
 $AP = OP = \frac{3}{2} \text{ units}$ $\perp$

$\therefore \triangle ADP \equiv \triangle COP$ (AAS)

(f) Join B to P
 $\therefore \triangle BDP \equiv \triangle BOP$ (RHS) $\perp$

Area of Quad DPOB
 $= 2 \times \triangle POB$
 $= 2 \times \frac{1}{2} \times 3 \times \frac{3}{2}$ $\perp$
 $= 4\frac{1}{2} \text{ square units}$

QUESTION 4 (12 MARKS)



Using Pythagoras theorem

$$h = \sqrt{13^2 - 12^2}$$

$$= \sqrt{25} = 5 \text{ units}$$

$$\therefore \tan \theta = \frac{5}{12}$$

(b) $2 \sin \theta = -1$

$$\sin \theta = -\frac{1}{2}$$

basic angle: 30°

$\therefore$ Sin -ve in 3rd Quad:

$$\theta = 210^\circ$$

Sin -ve in 4th Quad:

$$\theta = 330^\circ$$

(c) $\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x}$

$$= \frac{\sin^2 x + (1 + \cos x)^2}{\sin x (1 + \cos x)}$$

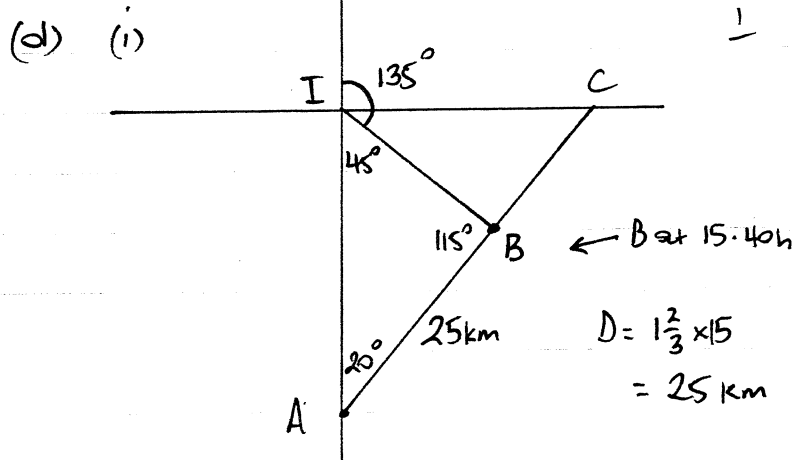
$$= \frac{\sin^2 x + 1 + 2\cos x + \cos^2 x}{\sin x (1 + \cos x)}$$

$$= \frac{2 + 2\cos x}{\sin x (1 + \cos x)}$$

$$= \frac{2(1 + \cos x)}{\sin x (1 + \cos x)}$$

$$= \frac{2}{\sin x}$$

$$= 2 \operatorname{cosec} x$$



(ii) $\frac{IA}{\sin 115^\circ} = \frac{25}{\sin 45^\circ}$

$$IA = \frac{25}{\sin 45^\circ} \times \sin 115^\circ$$

$$= 32.0428 \dots$$

$$\approx 32 \text{ km (to 2 sig. fig.)}$$

(iii) Find AC to find time from A to C

$$\cos 20^\circ = \frac{IA}{AC}$$

$$AC = \frac{32}{\cos 20^\circ} = 34.1 \text{ km}$$

$$\text{Time} = \frac{\text{Dist}}{\text{speed}} = \frac{34.1 \text{ km}}{15 \text{ km/h}}$$

$$= 2.27 \text{ h}$$

$$= 2 \text{ h } 16 \text{ min}$$

$\therefore$ Leaving at 14.00 h, ship

will be east of island

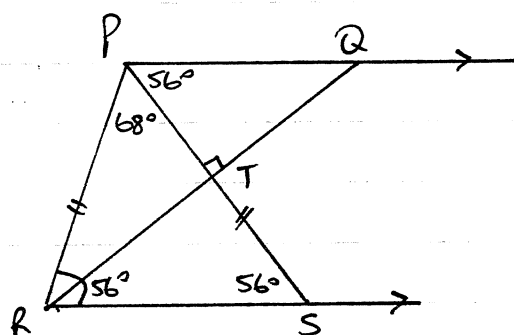
at 14h + 2h 16 min

= 16h 16 min

QUESTION 5

(8 MARKS)

(2)



(i) $\angle PRS = 56^\circ$ 1

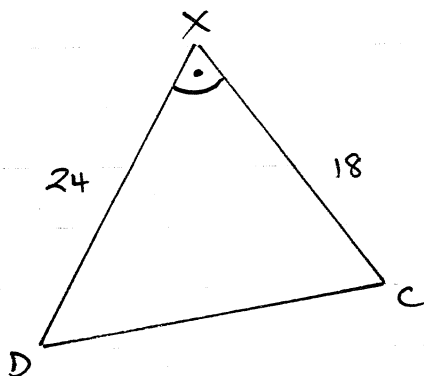
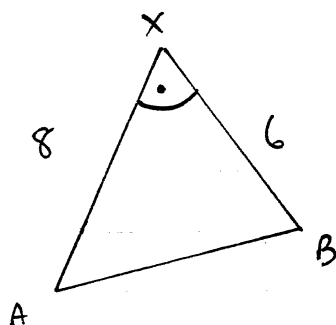
(ii) $\angle RPS = 68^\circ$ ($\angle$ sum of isos. $\triangle PRS$) 1

$\therefore \angle PRQ = 22^\circ$ ($\angle$ sum of $\triangle PRT$) 1

(b) (i) Angle sum $= 6 \times 180^\circ = 1080^\circ$ 1

(ii) Size of interior $\angle = \frac{1080^\circ}{8} = 135^\circ$ 1

(c)



$\angle X$ is the common angle 1

$$\frac{XB}{XA} = \frac{6}{8} = \frac{3}{4}$$

1

$$\frac{XC}{XD} = \frac{18}{24} = \frac{3}{4}$$

$\therefore \triangle XAB \parallel \triangle XDC$ because the corresponding sides 1
around the common angle are in the same ratio.

QUESTION 6

(16 MARKS)

$$(2) \quad (i) \quad \frac{dy}{dx} = 6x^2 + 5 + x^{-2} \quad \downarrow$$

$$= 6x^2 + 5 + \frac{1}{x^2} \quad \downarrow$$

$$(ii) \quad \frac{dy}{dx} = 3(1-4x^5)^2 \cdot -20x^4 \quad \downarrow$$

$$= -60x^4(1-4x^5)^2 \quad \downarrow$$

$$(iii) \quad \frac{dy}{dx} = 4x \cdot \frac{1}{2}x^{-\frac{1}{2}} + \sqrt{x} \cdot 4 \quad \downarrow$$

$$= \frac{2x}{\sqrt{x}} + 4\sqrt{x} \quad \downarrow$$

$$= 2\sqrt{x} + 4\sqrt{x} \quad \downarrow$$

$$= 6\sqrt{x} \quad \downarrow$$

$$(iv) \quad \frac{dy}{dx} = \frac{(1-2x) \cdot 6 - (6x+5) \cdot 2}{(1-2x)^2} \quad \downarrow$$

$$= \frac{6-12x+12x+10}{(1-2x)^2} \quad \downarrow$$

$$= \frac{16}{(1-2x)^2} \quad \downarrow$$

$$(b) \quad \lim_{x \rightarrow \infty} \frac{\frac{4x^2}{x^2} + \frac{x}{x^2}}{\frac{1}{x^2} + \frac{2x}{x^2} - \frac{x^2}{x^2}} \quad \downarrow$$

$$= \frac{4}{-1} = -4 \quad \downarrow$$

$$(c) \quad (i) \quad \frac{dv}{dt} = 8t - 12 \quad \downarrow$$

$$(ii) \quad 8t - 12 = 0$$

$$8t = 12$$

$$t = \frac{12}{8} = 1\frac{1}{2} \quad \downarrow$$

$$(d) \quad \text{when } x=3, y = \frac{1}{2}$$

$$\therefore \text{Point is } (3, \frac{1}{2})$$

$$\text{grad} = \frac{(x+1) \cdot 0 - 2 \cdot 1}{(x+1)^2}$$

$$= \frac{-2}{(x+1)^2} \quad \downarrow$$

$$\text{at } x=3, \text{ gradient} = \frac{-2}{(3+1)^2} \quad \downarrow$$

$$= \frac{-2}{16} = -\frac{1}{8}$$

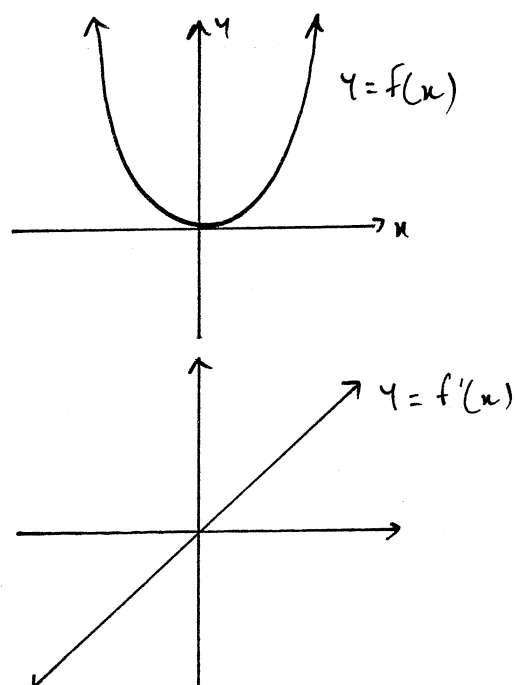
$$\text{Eqn of tang: grad} = -\frac{1}{8}, (3, \frac{1}{2})$$

$$\frac{y - \frac{1}{2}}{x - 3} = -\frac{1}{8}$$

$$8y - 4 = -x + 3 \quad \downarrow$$

$$x + 8y - 7 = 0$$

(e)



QUESTION 7 (8 marks)

(a) $y = -2x^2 + 7x + 4$

(i) $-2x^2 + 7x + 4 = 0$

$2x^2 - 7x - 4 = 0$

$\frac{(2x-8)(2x+1)}{2} = 0$

$(x-4)(2x+1) = 0$

$\therefore$ Roots are 4 and $-\frac{1}{2}$

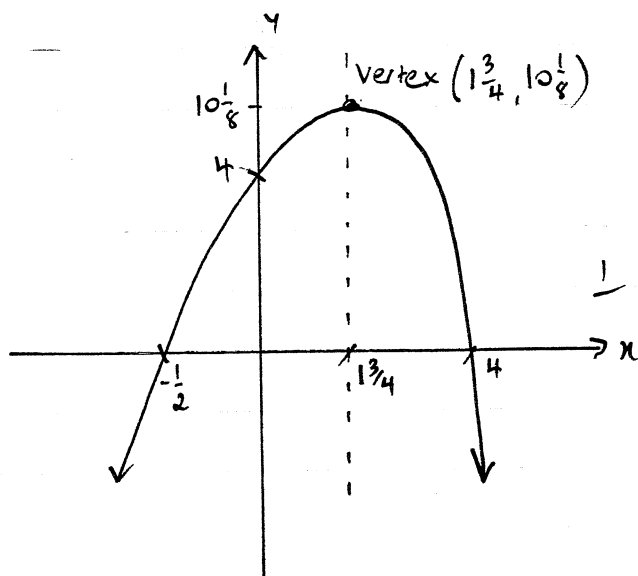
(ii) $x = \frac{-7}{2 \times -2} = \frac{-7}{-4}$

Axis of symmetry $x = 1\frac{3}{4}$

(iii) $y = -2\left(1\frac{3}{4}\right)^2 + 7 \times 1\frac{3}{4} + 4$
 $= 10\frac{1}{8}$

$\therefore$ Vertex is $\left(1\frac{3}{4}, 10\frac{1}{8}\right)$

(iv)



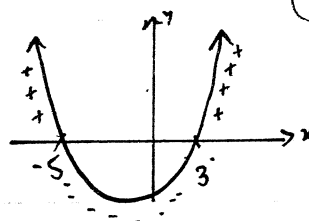
(b) No real roots $\Delta < 0$

$(k+1)^2 - 4 \cdot 1 \cdot 4 < 0$

$k^2 + 2k + 1 - 16 < 0$

$k^2 + 2k - 15 < 0$

$(k+5)(k-3) < 0$



$\therefore x^2 + (k+1)x + 4 = 0$

has no real roots

when

$-5 \leq k \leq 3$